Exercises

Exercise 5.1 · Collinearity and orthogonality. Consider the points A, B, C, D with affixes $a = 1+i$, $b = 3+2i$, $c = -1$ and $d = 2-i$.

1. Compute $\dfrac{c-a}{b-a}$. What can be deduced?
2. Compute $\dfrac{d-a}{b-a}$. What can be deduced about the triangle ABD?

Solution.

1. $\dfrac{-2-i}{2+i} = -1$, real: A, B, C are collinear (A is even the midpoint of $[BC]$).
2. $\dfrac{1-2i}{2+i} = \dfrac{(1-2i)(2-i)}{5} = \dfrac{-5i}{5} = -i$: $AD = AB$ and $(AB) \perp (AD)$. The triangle ABD is right-angled and isosceles at A.

Exercise 5.2 · Nature of triangles. Determine the nature of the triangle ABC in each case:

1. $a = 0$, $b = 2$, $c = 1 + i\sqrt{3}$;
2. $a = 1$, $b = 2+i$, $c = i$.

Solution.

1. $\dfrac{c-a}{b-a} = \dfrac{1+i\sqrt{3}}{2} = \mathrm{e}^{i\pi/3}$: equilateral triangle.
2. $\dfrac{c-a}{b-a} = \dfrac{-1+i}{1+i} = \dfrac{(-1+i)(1-i)}{2} = \dfrac{2i}{2} = i$: triangle right-angled and isosceles at A.

Exercise 5.3 · Sixth roots. Determine the elements of $\mathbb{U}_6$ in exponential form, then in algebraic form. Check that their sum is zero.

Solution. $\mathrm{e}^{ik\pi/3}$ for $k \in \{0, \dots, 5\}$: 1, $\frac{1}{2} + \frac{\sqrt{3}}{2}i$, $-\frac{1}{2} + \frac{\sqrt{3}}{2}i$, -1, $-\frac{1}{2} - \frac{\sqrt{3}}{2}i$, $\frac{1}{2} - \frac{\sqrt{3}}{2}i$. The real parts add up to $1 + \frac{1}{2} - \frac{1}{2} - 1 - \frac{1}{2} + \frac{1}{2} = 0$, and the imaginary parts cancel in pairs.

Exercise 5.4 · Cube roots of a complex number. Solve in $\mathbb{C}$ the equation $z^3 = 8i$. Write $z = r\mathrm{e}^{i\theta}$ and give the solutions in algebraic form.

Solution. $8i = 8\mathrm{e}^{i\pi/2}$. The equation is equivalent to $r^3 = 8$ and $3\theta \equiv \frac{\pi}{2} \pmod{2\pi}$, that is $r = 2$ and $\theta = \frac{\pi}{6} + \frac{2k\pi}{3}$. The solutions are $2\mathrm{e}^{i\pi/6} = \sqrt{3} + i$, $2\mathrm{e}^{5i\pi/6} = -\sqrt{3} + i$ and $2\mathrm{e}^{3i\pi/2} = -2i$. Check: $(-2i)^3 = -8i^3 = 8i$.

Exercise 5.5 · Completing a square. Let $A(1+2i)$ and $B(4+3i)$. Determine the affixes of the points C and D such that $ABCD$ is a direct square. Check that the diagonals have the same midpoint.

Solution. $b-a=3+i$ and $i(b-a)=-1+3i$. So $c=b+i(b-a)=3+6i$ and $d=a+i(b-a)=5i$. The midpoint of $[AC]$ has affix $\frac{4+8i}{2}=2+4i$, as does that of $[BD]$.

Exercise 5.6 · Sets of points. Let A and B be the points with affixes $2i$ and -1. To every point M with affix $z\neq -1$, we associate $Z=\dfrac{z-2i}{z+1}$.

1. Interpret geometrically the modulus and an argument of Z when $z\neq 2i$.
2. Determine the set of points M such that Z is real.
3. Determine the set of points M such that Z is purely imaginary and nonzero.
4. Determine the set of points M such that $|Z|=1$.

Solution.

1. $|Z|=\dfrac{MA}{MB}$ and $\arg Z\equiv\left(\overrightarrow{MB},\overrightarrow{MA}\right)\pmod{2\pi}$.
2. $Z=0$ for $M=A$. Otherwise, Z real and nonzero is equivalent to $\overrightarrow{MB}$ and $\overrightarrow{MA}$ collinear, that is $M\in(AB)$. The set is the line (AB) without B.
3. Z purely imaginary and nonzero is equivalent to $M\neq A$, $M\neq B$ and $(MA)\perp(MB)$: it is the circle with diameter $[AB]$ without A and B. Its centre has affix $\frac{-1+2i}{2}$ and its radius is $\frac{\sqrt{5}}{2}$.
4. $MA=MB$: the perpendicular bisector of $[AB]$.

Exercise 5.7 · A sum of cosines.

1. Let $n\geq 2$. Deduce from the course that $\displaystyle\sum_{k=0}^{n-1}\cos\frac{2k\pi}{n}=0$.
2. Write this equality for $n=5$ and prove that $1+2\cos\frac{2\pi}{5}+2\cos\frac{4\pi}{5}=0$.

Solution.

1. The sum of the elements of $\mathbb{U}_n$ is zero; so is its real part.
2. $\cos\frac{6\pi}{5}=\cos\left(2\pi-\frac{4\pi}{5}\right)=\cos\frac{4\pi}{5}$ and $\cos\frac{8\pi}{5}=\cos\frac{2\pi}{5}$. The sum $\cos 0+\cos\frac{2\pi}{5}+\cos\frac{4\pi}{5}+\cos\frac{6\pi}{5}+\cos\frac{8\pi}{5}$ therefore equals $1+2\cos\frac{2\pi}{5}+2\cos\frac{4\pi}{5}=0$.

Problem

Problem · Constructing a regular pentagon with ruler and compass. Let $\omega = e^{2i\pi/5}$ and $u = \omega + \frac{1}{\omega}$.

1. Prove that $u = 2\cos\frac{2\pi}{5}$.
2. Prove that $1 + \omega + \omega^2 + \omega^3 + \omega^4 = 0$, then, dividing by ω^2, that $u^2 + u - 1 = 0$.
3. Deduce that $\cos\frac{2\pi}{5} = \frac{\sqrt{5}-1}{4}$.
4. In an orthonormal frame, draw the unit circle and place $I(-\frac{1}{2}, 0)$ and $J(0, 1)$.
   (a) Compute IJ.
   (b) The circle with centre I through J meets the positive half of the horizontal axis at K. Prove that the abscissa of K is $2\cos\frac{2\pi}{5}$.
   (c) Deduce a ruler-and-compass construction of the vertices of a regular pentagon inscribed in the unit circle.

Solution · Problem.

1. $|\omega| = 1$, so $\frac{1}{\omega} = \overline{\omega}$ and $u = \omega + \overline{\omega} = 2\,\mathrm{Re}(\omega) = 2\cos\frac{2\pi}{5}$.
2. It is the sum of the fifth roots of unity. Dividing by $\omega^2 \neq 0$: $\omega^{-2} + \omega^{-1} + 1 + \omega + \omega^2 = 0$. Now $u^2 = \omega^2 + 2 + \omega^{-2}$, so $u^2 + u - 1 = \left(\omega^2 + \omega^{-2}\right) + \left(\omega + \omega^{-1}\right) + 1 = 0$.
3. The roots of $X^2 + X - 1$ are $\frac{-1\pm\sqrt{5}}{2}$. Since $0 < \frac{2\pi}{5} < \frac{\pi}{2}$, $u > 0$: $u = \frac{\sqrt{5}-1}{2}$ and $\cos\frac{2\pi}{5} = \frac{\sqrt{5}-1}{4}$.
4. (a) $IJ = \sqrt{\frac{1}{4} + 1} = \frac{\sqrt{5}}{2}$.
   (b) K has abscissa $-\frac{1}{2} + \frac{\sqrt{5}}{2} = \frac{\sqrt{5}-1}{2} = u$.
   (c) Construct K, then the midpoint H of $[OK]$, whose abscissa is $\cos\frac{2\pi}{5}$. The perpendicular to the axis at H meets the unit circle at ω and $\overline{\omega} = \omega^4$. Marking off the length $|\omega - 1|$ with the compass from these points gives ω^2 and ω^3.

Teacher note · Status of the problem. The trigonometric values of $\frac{2\pi}{5}$ and the construction of the pentagon are a "possible problem" of the syllabus. Have the construction carried out with dynamic geometry software, then check that the five sides have the same length ($|\omega - 1| = 2\sin\frac{\pi}{5}$).